Document 2: Credit System & Course Registration

Institution: Hyderabad Institute of Technology (HIT)

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1. Choice Based Credit System (CBCS) Architecture

Hyderabad Institute of Technology (HIT) operates strictly under the Choice Based Credit System (CBCS) model. This framework treats education as a modular learning experience, allowing students to navigate their academic pathways based on individual career goals, pacing preferences, and interdisciplinary interests.

1.1 Core Mathematical Definition of a Credit

At HIT, academic credits are mathematically mapped to direct student workload and contact hours per week over a standard 16-week semester framework. The calculation rules are defined as follows:

- **1 Lecture (L) Hour per week** = 1 Credit
- **1 Tutorial (T) Hour per week** = 1 Credit
- **2 Practical/Laboratory (P) Hours per week** = 1 Credit
- **3 Project Work/Field Internship Hours per week** = 1 Credit

Mathematical Equation for Course Credits (C):

$$C = L + T + P/2$$

1.2 Total Graduation Credit Thresholds

To be awarded the Bachelor of Technology (B.Tech) degree from HIT, a candidate must successfully complete all mandatory coursework and cross the minimum aggregate credit threshold.

- **Standard Minimum Graduation Requirement:** 160 Total Credits.
- **Lateral Entry Student Requirement:** 122 Total Credits (Admitted directly into the 3rd semester / 2nd year).

- **Maximum Permissible Registration Cap:** A student cannot register for more than 24 credits in any standard semester.
- **Minimum Absolute Registration Floor:** A student must register for at least 16 credits in a standard semester to maintain active student enrollment status.

2. Comprehensive Course Categorization and Weightage

The 160 credits required for graduation are split into 7 distinct categories. This ensures that while students have the flexibility to choose electives, they still build a rock-solid foundation in core engineering principles.

2.1 Detailed Credit Distribution Table

The following matrix dictates the exact minimum credit weights required across all categories:

Category Code	Course Category Description	Core Focus Areas	Minimum Credits Required
BSC	Basic Science Courses	Mathematics, Physics, Chemistry	24
ESC	Engineering Science Courses	Basic Electrical, Workshop, Coding Fundamentals	20
PCC	Professional Core Courses	Branch-Specific Mandatory Advanced Subjects	48
PEC	Professional Elective Courses	Specialized Tracks within the Parent Branch	18

OEC	Open Elective Courses	Cross-Disciplinary Subjects from Other Branches	12
PROJ	Project Work & Internships	Industry Internships, Mini Projects, Major Thesis	24
MC	Mandatory Non-Credit Courses	Environmental Studies, Constitution of India	0 (Pass Required)

2.2 Mandatory Non-Credit Course Regulations (Category MC)

Courses falling under Category MC do not contribute numerical points to the semester GPA or cumulative CGPA calculation. However, they are a strict legal prerequisite for the degree award.

- Students must maintain a minimum of 75% attendance in MC courses.
- Evaluation is conducted via a 50-mark internal written exam at the end of the semester.
- The final result is recorded on the grade sheet simply as **P (Pass)** or **NP (Not Passed)**. A minimum of 40% marks is required to secure a Pass.
- If a student receives an **NP** grade, they must re-register and re-appear for the exam in the subsequent semester.

3. Semester-Wise Standard Credit Distribution Path

To ensure students graduate within the standard 4-year timeframe, the university recommends following a highly optimized semester-wise structural timeline.

3.1 First Year (Freshman Year Hierarchy)

- **Semester 1:** 21 Credits (Focuses primarily on BSC and ESC courses, including Calculus, Engineering Physics, and Programming in Python).
- **Semester 2:** 19 Credits (Includes Linear Algebra, Engineering Chemistry, Data Structures, and Basic Electrical Engineering).

3.2 Second Year (Sophomore Year Hierarchy)

- **Semester 3:** 22 Credits (Transition into parent department core subjects. First introduction to Professional Core Courses - PCC).
- **Semester 4:** 20 Credits (Advanced core programming, discrete mathematics, and initial introduction to minor industry projects).

3.3 Third Year (Junior Year Hierarchy)

- **Semester 5:** 22 Credits (Heavy concentration of branch core subjects along with the selection of the first Professional Elective - PEC-1).
- **Semester 6:** 20 Credits (Advanced electives, introduction to the first cross-disciplinary Open Elective - OEC-1, and mandatory Mini Project submission).

3.4 Fourth Year (Senior Year Hierarchy)

- **Semester 7:** 20 Credits (Advanced specialized electives, OEC-2, OEC-3, and initial phase initialization of the Major Graduation Thesis).
- **Semester 8:** 16 Credits (Reserved almost entirely for a full-semester Industry Internship or full-time Major Project execution).

4. The Formal Course Registration Workflow

Course registration is an administrative binding process. No student is allowed to attend classes, sit in laboratories, or take internal assessments for a course unless they have formally registered for it through the HIT Student Portal.

4.1 Step-by-Step Registration Timeline

The registration window opens systematically before the start of each academic term:

1. **Phase 1: Academic Advising (Weeks -4 to -2):** Every student is assigned a Faculty Advisor. The student must meet with their advisor to review their past CGPA performance, identify outstanding backlogs, and map out their choices for PEC and OEC classes.
2. **Phase 2: Slot Allocation Pre-Registration (Weeks -2 to -1):** Elective slots are allocated based on a first-come, first-served basis combined with a CGPA merit filter. High-demand courses cap out at 60 seats per section.
3. **Phase 3: Final Enrollment (Days 1 to 3 of the new semester):** Students log into the portal, clear any outstanding financial dues with the accounts department, freeze their final timetables, and print their official Registration Slip.
4. **Phase 4: Late Registration Window (Days 4 to 10):** Students who missed the final enrollment deadline may register with a strict monetary fine penalty of ₹500 per day. The portal locks completely at midnight on Day 10. No registrations can be processed after this point under any circumstances.

4.2 Strict Prerequisite Mapping Matrix

Advanced engineering courses at HIT are locked behind structural prerequisites to ensure foundational mastery. If a student has an active backlog or failed a prerequisite course, the portal will completely block them from selecting the advanced course.

Target Course Code & Title	**Prerequisite Course Required**	**Minimum Prerequisite Status**
AIDS-302: Advanced Machine Learning	**CS-201:** Data Structures and Algorithms	Clear Pass (Grade $\ge$ 5.0)
CS-401: Distributed Cloud Architecture	**CS-304:** Operating Systems Fundamentals	Clear Pass (Grade $\ge$ 5.0)
ECE-305: Digital Signal Processing	**ECE-202:** Signals and Systems	Clear Pass (Grade $\ge$ 5.0)
MEC-402: Advanced Fluid Dynamics	**MEC-203:** Thermodynamics and Fluids	Clear Pass (Grade $\ge$ 5.0)

5. Course Modification and Withdrawal Policies

HIT recognizes that students may sometimes need to adjust their academic workloads after a semester has begun.

5.1 The Standard Add/Drop Window

Students can modify their elective schedules during the first 14 calendar days of a standard semester.

- **Permissible Changes:** A student can drop a Professional Elective or Open Elective and add an alternative, provided there are open slots available in the target course section.
- **Core Course Restriction:** Professional Core Courses (PCC) cannot be added or dropped under any circumstances. They are mandatory and statically fixed to that specific semester timeline.

5.2 Formal Course Withdrawal (W-Grade Policy)

If a student finds themselves struggling academically in an elective course mid-semester, they can opt for a formal academic withdrawal to protect their CGPA from an F-grade.

- **Deadline:** Withdrawal applications must be submitted to the Controller of Examinations before the end of Week 8 of the semester.
- **Academic Transcript Impact:** The course will not be completely erased from the student's records. Instead, it will be marked permanently on their transcript with a **W Grade (Withdrawal)**.
- **Credit Impact:** A "W" grade carries zero credits. The student must re-register for the course (or a completely different elective equivalent) in a future semester and pay the full standard course fee again.
- **Attendance Caveat:** A course withdrawal will only be approved if the student has maintained a minimum of 75% attendance up to the exact date of their withdrawal request.

6. Fast-Track Accelerated Summer Term Architecture

The Fast-Track Summer Term is a highly intensive, compressed 6-week academic session running through the annual vacation months of June and July. It is explicitly designed to act as an academic safety net for students struggling with backlogs.

6.1 Regulatory Constraints and Eligibility Rules

Due to the highly compressed nature of the summer term, strict limitations are placed on student choices:

- **Maximum Credit Allocation:** A student can register for a maximum of 3 subjects or an absolute limit of 12 credits total during the summer session.
- **Target Course Restrictions:** A student can *only* register for a course in the summer term if they have previously attempted and failed it (Grade F) or were officially detained from writing the main exam due to an attendance shortage.
- **First-Time Course Prohibitions:** Students are completely prohibited from taking a brand-new, first-time elective course during the summer term to try to accelerate their graduation timeline.

6.2 Financial Slabs and Fees for Summer Registration

Because the summer session runs outside the standard academic calendar, it requires independent financial clearance per course choice:

- **Standard Theory-Based Course:** ₹2,500 per credit. (e.g., A 3-credit math class costs ₹7,500).
- **Practical Laboratory-Based Course:** ₹3,500 per credit. (e.g., A 2-credit programming lab costs ₹7,000).
- **Summer Administrative Infrastructure Surcharge:** A flat fee of ₹1,500 is applied to every registering student regardless of the number of courses selected.

7. B.Tech Honors Degree Specialization Policy

For highly ambitious students who want to show exceptional expertise within their parent department, HIT offers a prestigious "B.Tech with Honors" degree track.

7.1 Strict Entry Gatekeeping Milestones

To qualify for enrollment into the Honors track, a student must meet these specific milestones by the end of their 4th semester:

- **Minimum CGPA Cutoff:** The student must maintain a minimum Cumulative GPA of 8.0 or higher continuously across semesters 1, 2, 3, and 4.
- **Backlog Cleared Status:** The student must have zero active backlogs and must never have received an F-grade or detention in any subject up to the application date.
- **Timing:** Applications must be submitted within 7 days of the announcement of the 4th-semester final results.

7.2 Extra Academic Load Requirements

Earning the Honors tag requires taking on a significantly heavier workload than the average engineering student:

- **Additional Credit Requirements:** The student must earn an additional 20 credits completely separate from the standard 160 credits required to graduate (Totaling 180 credits overall).
- **Course Assignment Level:** These 20 extra credits must be earned by passing advanced, master-level (M.Tech) courses or pre-approved, high-level online research certification paths (like NPTEL/SWAYAM advanced modules).
- **Parent Department Restriction:** All 20 credits must be earned from subjects within the student's parent department. For example, a CSE student must take advanced AI, cryptography, or compiler design courses. They cannot take mechanical or civil courses for Honors credits.

8. B.Tech Minor Degree Specialization Policy

To support interdisciplinary innovation and broaden career opportunities, HIT allows students to earn a "Minor Degree" in an engineering field completely outside of their parent department.

8.1 Gatekeeping and Entry Thresholds

The entry requirements for a Minor specialization are balanced to allow broad student participation while maintaining rigorous academic standards:

- **Minimum CGPA Cutoff:** The student must maintain a Cumulative GPA of 7.0 or higher continuously through their first two years of engineering study.
- **Backlog Status:** The student must have zero active backlogs at the time of entry in the 5th semester.

8.2 Structural Minor Coursework Layout

- **Additional Credit Requirements:** The student must complete an additional 18 credits beyond the standard graduation requirements.
- **Cross-Department Restriction:** These 18 credits must be chosen entirely from the fundamental core classes offered by a department other than the student's parent department.
- *Approved Minor Specialization Tracks:*
  - A Mechanical Engineering student can earn a **Minor in Computer Science** by taking 18 credits of coding, database management, and operating systems courses.
  - A CSE student can earn a **Minor in Robotics** by taking 18 credits of control systems and automation coursework from the Mechanical Engineering department.
  - An ECE student can earn a **Minor in Data Science** by taking 18 credits of statistical modeling and analytics modules from the AI & DS department.